PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: MedFlow AI assistance

Version: 1.0

1. Project Overview

- **Problem Statement:** Medical administrative workflows are currently fragmented and manual, relying on unstructured data like voice notes and handwritten forms that require high human coordination, leading to diagnostic delays and clinical errors.
- **Target Domain:** Intelligent medical administrative processes and automated patient triage systems.
- **Proposed Solution Summary:** MedFlow acts as a stateful, adaptive workflow engine using Z.AI's GLM as the central reasoning core to transform unstructured clinical inputs into structured medical profiles and automated action plans.

2. Background & Business Objective

- **Background of the Problem:** For years, patient intake has followed a manual, linear pathway where staff transcribe information, cross-check guidelines, and manually coordinate with insurance and scheduling systems, creating a technical and labor bottleneck.
- **Importance of Solving This Issue:** By accelerating triage and data entry from hours to seconds, clinics can test patient flow ideas rapidly and ensure critical cases are identified without the barrier of manual coordination.

- **Strategic Fit / Impact:** It aligns with the AI-Native Development ecosystem by integrating Z.AI GLM to facilitate a reliable, scalable, and intelligent healthcare workspace.

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3. Product Purpose

3.1. Main Goal of the System: The primary objective of MedFlow is to provide an AI-enabled agentic partner in clinical administration that autonomously handles the complex coordination of patient data, triage, and scheduling.

3.2. Intended Users (Target Audience):

- Clinical Staff and Nurses
- Doctors
- Hospital Administrators
- Health Clinic Founders
- Patient
- First Year Software Engineering Students

4. System Functionalities

4.1. Description: The system operates as a high speed generative reasoning engine for clinical intake through natural language, capturing medical relationships and multi-screen layout flows for healthcare workers.

4.2. Key Functionalities:

- **Vibe Triage**: Translates human language symptoms and descriptive instructions into high-priority clinical triage reports.
- **Conversational Editing:** Facilitates continuous modification of the patient's medical profile by healthcare workers through direct text or voice interaction.
- **Versioning & Infinite Canvas:** Features an AI-enabled "Reasoning Canvas" to track different diagnostic paths and agentic decisions throughout the patient journey
- **Export- Component Based:** Features an AI-enabled "Reasoning Canvas" to track different diagnostic paths and agentic decisions throughout the patient journey

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: The system utilizes Z.AI GLM (General Language Model) as the central reasoning engine. Our model was selected because it acts as a stateful and adaptive workflow engine, capable of transforming unstructured data into intelligent, automated processes. Its core capabilities include the understanding of unstructured inputs (messages/voice), multi-step reasoning, and dynamic task orchestration via tool or API interactions.

4.3.2. Prompting Strategy: Our team employs a multi-step agentic prompting strategy. This approach fits the medical use case because it allows the GLM to break down a complex medical intake into distinct, logical phases from Symptom Extraction to Clinical Reasoning to Actionable Output Generation. It ensures the system can handle real world constraints such as ambiguity or incomplete patient data by iteratively prompting for clarification when necessary.

4.3.3. Context & Input Handling: MedFlow handles unstructured medical inputs by extracting key clinical entities into a structured context window. Our system accepts a maximum input size of 1,500 tokens (approximately 800 words or 3 pages of text). Inputs exceeding this limit are managed through chunking into segments, ensuring the GLM processes the data without losing critical medical context.

4.3.4. Fallback & Failure Behavior: If the model returns off-topic or unusable responses, the system enters a graceful degradation state. To mitigate hallucinations, our system implements a human-in-the-loop escalation path, where a medical professional must approve structured outputs before they are finalized. In cases of API rate limitations or failures, our system uses automated retrying or triggers an Emergency Rollback to a stable "Golden Version" to ensure patient safety.

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5. User Stories & Use Cases

- **User Stories:**
  - "As a **Clinical Nurse**, I want to record a voice note of a patient's symptoms so that the system automatically generates a structured medical profile without me having to type it manually."
  - "As a **Doctor**, I want to see a visual reasoning path of how the AI determined a triage priority so that I can verify the logic before approving the treatment plan."
  - "As a **Hospital Administrator**, I want the system to automatically check insurance coverage and available slots so that the administrative coordination is handled without manual phone calls."

- **Use Cases (Main Interactions):**

  Prompt to Structured Profile Flow: Users provide unstructured input (text/voice). MedFlow parses the clinical intent, identifies symptoms and history, and renders a structured, editable profile on the dashboard.

  Agentic Triage Generation: The system analyzes the extracted symptoms against medical guidelines and triggers the "Next Action" (e.g., Generate Referral or Schedule Lab Test) while maintaining consistent clinical tokens like patient ID and severity level.

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6. Features Included (Scope Definition)

- A conversational and visual canvas for healthcare workers to monitor agentic reasoning and edit patient data in a 2D workspace.
- Coherent and consistent clinical flows that manage the entire journey from intake to final appointment scheduling.
- One-click export feature that converts the AI-generated triage report into a standardized PDF or hospital-ready code snippets.
- Support for both unstructured text-to-workflow and voice-to-workflow features to accommodate busy clinical environments.

7. Features Not Included (Scope Control)

- Implementation of full backend pharmacy logic.
- Only for frontend scaffolding, does not support production environment.
- Multi-user simultaneous editing are not supported [Initial Versions].

8. Assumptions & Constraints

- **LLM Cost Constraint:** Average token cost is estimated to be manageable by caching repeated clinical queries and truncating non-essential conversational fillers from patient voice notes.
- **Technical Constraints:** Code export and report generation are currently limited to HTML, CSS, and PDF formats; integration with legacy proprietary hospital software (EHR) is not supported in the MVP.
- **Performance Constraints:** To maintain high-quality clinical reasoning, there is a daily generation limit on complex triage paths per user session.
- **User Input:** The model acts as a copilot and **requires** human approval from a medical professional before any high-risk clinical action is finalized.

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9. Risks & Questions Throughout Development

- **Clinical Accuracy**: How do we ensure the AI doesn't hallucinate symptoms or provide conflicting triage advice across different patients?
- **State Stability:** Does the healthcare worker need to manually fix the structured profile if the GLM fails to parse a thick accent in a voice note?
- **Interpretation of Ambiguity**: What will happen if the AI is unable to contextualize vague symptoms like "feeling unwell" without specific data points?

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